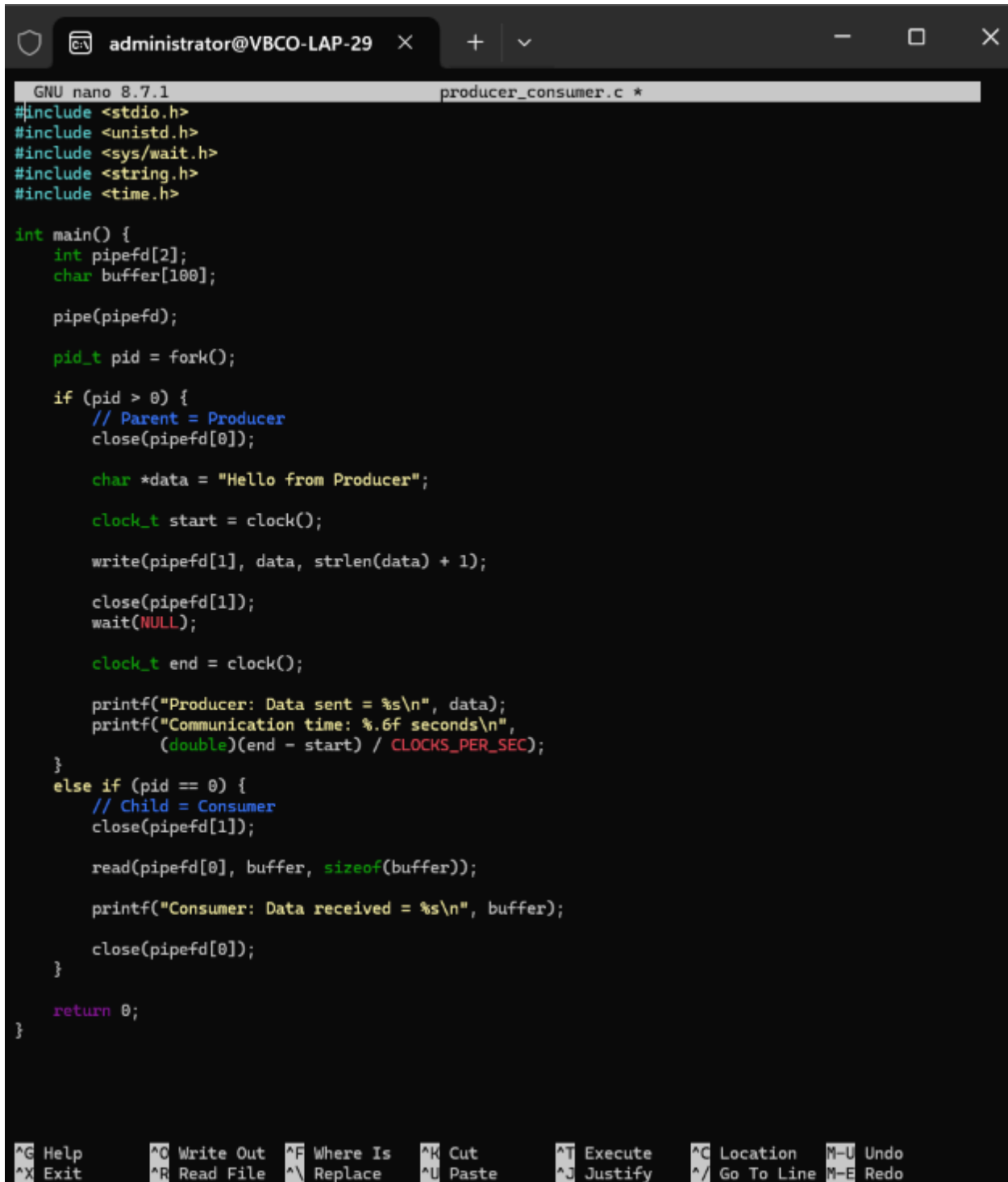


PRACTICAL-5

INPUT



```
GNU nano 8.7.1 producer_consumer.c *
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
#include <string.h>
#include <time.h>

int main() {
    int pipefd[2];
    char buffer[100];

    pipe(pipefd);

    pid_t pid = fork();

    if (pid > 0) {
        // Parent = Producer
        close(pipefd[0]);

        char *data = "Hello from Producer";

        clock_t start = clock();

        write(pipefd[1], data, strlen(data) + 1);

        close(pipefd[1]);
        wait(NULL);

        clock_t end = clock();

        printf("Producer: Data sent = %s\n", data);
        printf("Communication time: %.6f seconds\n",
            (double)(end - start) / CLOCKS_PER_SEC);
    }
    else if (pid == 0) {
        // Child = Consumer
        close(pipefd[1]);

        read(pipefd[0], buffer, sizeof(buffer));

        printf("Consumer: Data received = %s\n", buffer);

        close(pipefd[0]);
    }

    return 0;
}
```

^G Help ^O Write Out ^F Where Is ^K Cut ^I Execute ^C Location M-U Undo
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^_ Go To Line M-E Redo

OUTPUT

```
administrator@VBCO-LAP-29: ~  
administrator@VBCO-LAP-29:~$ nano producer_consumer.c  
administrator@VBCO-LAP-29:~$ gcc producer_consumer.c -o producer_consumer  
administrator@VBCO-LAP-29:~$ ./producer_consumer  
Consumer: Data received = Hello from Producer  
Producer: Data sent = Hello from Producer  
Communication time: 0.000077 seconds  
  
administrator@VBCO-LAP-29: ~  
administrator@VBCO-LAP-29:~$ ls -l | grep ".c"  
-rw----- 1 administrator administrator 892 Aug 29 09:28 p1.c.save  
-rw-r--r-- 1 administrator administrator 234 Aug 17 05:39 permissions.c  
drwxr-xr-x 2 administrator administrator 4096 Aug 13 07:09 practical-4  
-rw----- 1 administrator administrator 6290 Aug 25 08:21 practical.c.save  
-rw-r--r-- 1 administrator administrator 858 Aug 5 16:51 practical02.c  
-rw-r--r-- 1 administrator administrator 529 Aug 3 16:46 pro1.c  
-rwxr-xr-x 1 administrator administrator 16352 Aug 29 09:29 producer_consumer  
-rw-r--r-- 1 administrator administrator 893 Aug 29 09:35 producer_consumer.c  
-rw-r--r-- 1 administrator administrator 894 Aug 29 09:36 producer_consumer.c.save  
-rw-r--r-- 1 administrator administrator 137 Jul 29 08:03 program1.c
```